Linrui (Stephen) Tan

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EDUCATION

Rice University **Houston, TX**

Artificial Intelligence (Doctor of Philosophy) Anticipated Graduation: May 2030

- ✓ GPA: 4.00 out of 4.00
- ✓ Selected **Coursework**: AI for health, Machining learning, Deep Learning

University of Rochester **Rochester, NY**

Optical Engineering (Bachelor of Science) Anticipated Graduation: May 2025

- ✓ GPA: 3.97 out of 4.00

RESEARCH EXPERIENCES

Rice University **August 2025 - Present**

Research Assistant in computational wellbeing lab

- ✓ Previous mental health AI systems often relied on single-modality data (text-only or behavioral-only), which struggled to model stress accurately.
- ✓ This research introduces a multi-agent workflow combining a classical text classifier (Input Agent) with a fine-tuned LLM (Judge Agent) to fuse textual and behavioral data, resulting in statistically significant improvements.
- ✓ The project required multidisciplinary expertise in NLP, classical ML, data fusion, and human–AI interaction design, along with skills in fine-tuning LLMs, implementing interpretability tools (SHAP).

Boston University **June 2024 - August 2024**

Research Assistant (REU) at High Dimensional Photonics Lab

- ✓ One of the 12 attendees among 1200 applicants (***paper under review***).
- ✓ Current technology, including avalanche photodiodes (PDs), photomultiplier tubes (PMTs), and amplified photodiodes cannot simultaneously provide low-photon-flux sensitivity and high bandwidth operation due to limitations in speed, sensitivity, or dynamic range.
- ✓ Paired low-noise and high-gain parametric amplification in optical fibers with detection using conventional, high-bandwidth photodiodes.
- ✓ Observed a >10x increase in sensitivity at bandwidths >10x larger than conventional technologies allowing us to resolve signals as low as 20 photons at 1.6 GHz bandwidth. This method is drop-in compatible with any light detection system.

University of Rochester **January 2024 – May 2025**

Research Assistant at Laboratory of Nanophotonics

- ✓ Investigating LiDAR scanning by assembling the optical systems and the computer algorithms (mainly MATLAB code) for scanning the depth of focus of a 3D object built by LEGO, which could potentially apply to multiple applications on proximity tests.
- ✓ Testing and analyzing Lithium Niobate (LiNbO3) photonic chips aim to explore the nonlinear properties of the LiNbO3, focusing specifically on its performance in micro-ring resonator structures.
- ✓ Optimizing color coverage in EOM to find voltage (V_π) required to induce a π phase difference between the two modulator arms for 450 nm (blue), 520 nm (green), and 633 nm (red). The objective is to achieve a precise and wide color gamut for potential applications in a high-definition color spectrum.

Zhejiang University **July 2023 –August 2023**

Laboratory of Visual Intelligence and Pattern Analysis

- ✓ Explored the most popular AI model—diffusion model, and its usage.
- ✓ Worked on the improvement of the diffusion model by importing different contrast images for simulation. That feeds the AI system enough data for generating new images.

South China University of Technology **July 2021 - June 2022**

Chemical Engineering Laboratory Research

- ✓ Collaborated with the PhD students of SCUT to do experiments on interfacial catalysis.

- ✓ Learned to use laboratory equipment, such as high-speed centrifuge, gas chromatograph, and field emission scanning electron microscope (FESEM)
- ✓ Engaged in experiment design, observation, data analysis, and paper writing.

INTERNSHIP

ZEISS, CHINA **MAY 2023 – June 2023**

Research Assistant

- ✓ Finding the best method of measuring refractive index. Specifically, using python based on a mechanical control system at Nankai University, I tested different special types of glass (high refractive index, high diopter glasses and so on) and optimized the system during that process.
- ✓ Reducing the stress on the glasses' frame: apply rubber-like materials on the edge of the glasses' frames.

TEACHING ASSITANT EXPERIENCE

University of Rochester **January 2023 – May 2025**

Teaching Assistants

- ✓ Matlab TA --- MATLAB applications on optical engineering and Fourier optics in MATLAB
- ✓ Geometric Optics Workshop Leader --- Traditional ray tracing lens systems and modern lens design software.
- ✓ Electronics and Circuit TA --- Electronics components and circuit analysis
- ✓ Aberration theory TA --- Electromagnetic waves aberration and interferometers analysis.

VOLUNTEER

THE AGGY PROJECT **August 2024 – May 2025**

Advisors

- ✓ Sharing the educational pathway, discussing the positions, and providing high school students with advice.
- ✓ Aiding them with writing a research paper (literature review). The literature review is the central outcome and students will write literature reviews on the application of detectors concerning environmental pollutants

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Buddhist Student Association **May 2024 – May 2025**

Event Chair

- ✓ Designing multiple events (dharma studies, meditation sessions, and temple trips) for students to understand how to deal with pressure in the current fast-paced society through meditation, loving-kindness, and compassion.
- ✓ Prepared for student organization collaborations between Interfaith 101 Introduction to Religion Session, Student Philosophy Council, Hinduism Student Association, and Pagan Student Association.
- ✓ Sharing understanding of the vast religions, how people's belief system works, and the mechanism behind the belief and mental health from a Buddhist's perspective.

SKILLS

Communication, analytic thinking, Teaching ability, and etc.
Problem Solving Skills, Data Analysis, Laboratory Experience, Application of AI.
Proficient in MATLAB, Python, Image J, CodeV, Zemax, LaTeX